

Financial Econometrics - Exercise 3.23

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Exercise 3.23: Housing Prices in Collegetown (Quadratic Model)

This analysis utilizes a quadratic regression model to examine the relationship between house prices (PRICE) and their interior area (SQFT). The model is specified as:

$$PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$$

Part (a): Marginal Effect at 2,000 Square Feet

Answer:

- **Hypothesis:** We test the Null Hypothesis (H_0 : Marginal Effect ≤ 13) against the Alternative Hypothesis (H_1 : Marginal Effect > 13) for a 2,000 sqft house.
- **Marginal Effect Formula:** In this quadratic model, the marginal effect is $2\alpha_2 SQFT$. At 2,000 sqft ($SQFT = 20$), the marginal effect is $40\alpha_2$.
- **Conclusion:** Based on the calculated t-statistic and p-value, we determine whether the data provides sufficient evidence to conclude that an additional 100 square feet increases the expected price by more than \$13,000.

```
## a
```

```
# 1. 建立二次模型
```

```
# PRICE = α1 + α2SQFT^2 + e
```

```
mod_quadratic <- lm(price ~ I(sqft^2), data = collegetown)
```

```
# I(sqft^2) 做一般的數學運算，避免做成交互作用
```

```
result <- summary(mod_quadratic)
```

```
result
```

```
##
## Call:
## lm(formula = price ~ I(sqft^2), data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -383.67  -48.39   -7.50   38.75  469.70
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 93.565854   6.072226   15.41  <2e-16 ***
## I(sqft^2)    0.184519   0.005256   35.11  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 92.08 on 498 degrees of freedom
## Multiple R-squared:  0.7122, Adjusted R-squared:  0.7117
## F-statistic: 1233 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
# 3. 提取 alpha2 估計值與標準誤
# coef() 抓係數, vcov() 抓變異數矩陣
a2_hat <- coef(mod_quadratic)["I(sqft^2)"]
se_a2  <- sqrt(vcov(mod_quadratic)["I(sqft^2)", "I(sqft^2)"])

# 4. 計算 2000 sqft (sqft=20) 時的邊際效應 (ME = 2 * a2 * 20 = 40 * a2)
me_observed <- 40 * a2_hat
claim_value <- 13

# 5. 計算t檢定值 (基於邊際效應)
t_stat <- (me_observed - claim_value) / (40 * se_a2)

# 6. 提取自由度與計算 p-value
df_quadratic <- df.residual(mod_quadratic)
p_value <- 1 - pt(t_stat, df_quadratic)
t_critical <- qt(0.95, df_quadratic)

# 7. 輸出結果
cat("Marginal Effect at 2000 sqft: ", round(me_observed, 4), "\n")
```

```
## Marginal Effect at 2000 sqft:  7.3808
```

```
cat("Test Statistic (t): ", round(t_stat, 4), "\n")
```

```
## Test Statistic (t):  -26.7288
```

```
cat("P-value: ", round(p_value, 6), "\n")
```

```
## P-value:  1
```

```
cat("Critical Value (tc): ", round(t_critical, 4), "\n")
```

```
## Critical Value (tc): 1.6479
```

```
# 8. 結論判斷
```

```
if (t_stat > t_critical) {  
  cat("Conclusion: Reject H0. The marginal effect is significantly greater than $13,000.\n")  
} else {  
  cat("Conclusion: Do not reject H0. No evidence that marginal effect exceeds $13,000.\n")  
}
```

```
## Conclusion: Do not reject H0. No evidence that marginal effect exceeds $13,000.
```

Part (b): Marginal Effect at 4,000 Square Feet

Answer: * **Hypothesis:** We test H_0 : Marginal Effect ≤ 13 versus H_1 : Marginal Effect > 13 for a 4,000 sqft house. * **Marginal Effect Formula:** At 4,000 sqft ($SQFT = 40$), the marginal effect is $2\alpha_2 \times 40 = 80\alpha_2$. * **Economic Insight:** Because the model is quadratic, the marginal effect increases as the house size increases. This test examines whether the impact of size on price is significantly higher for larger properties compared to the \$13,000 threshold.

```
## b
```

```
# 1. 計算 Marginal Effect at 4000 sqft (SQFT = 40)
```

```
# (ME = 2 * a2 * 40 = 80 * a2)
```

```
# 使用 me_40 變數名，區隔不同坪數
```

```
me_40 <- 80 * a2_hat
```

```
claim_value <- 13
```

```
# 2. 計算t檢定值 (Test Statistic)
```

```
t_stat_b <- (me_40 - claim_value) / (80 * se_a2) # 給自己的提醒：標準誤須乘上倍數 80
```

```
# 3. 提取自由度與計算 p-value (單尾右側)
```

```
df_quadratic <- df.residual(mod_quadratic)
```

```
p_value_b <- 1 - pt(t_stat_b, df_quadratic)
```

```
t_critical_b <- qt(0.95, df_quadratic) # 單尾 5% 臨界值
```

```
# 4. 顯示結果
```

```
cat("Marginal Effect at 4000 sqft: ", round(me_40, 4), "\n")
```

```
## Marginal Effect at 4000 sqft: 14.7615
```

```
cat("Test Statistic (t): ", round(t_stat_b, 4), "\n")
```

```
## Test Statistic (t): 4.1895
```

```
cat("P-value: ", round(p_value_b, 6), "\n")
```

```
## P-value: 1.7e-05
```

```
cat("Critical Value (tc): ", round(t_critical_b, 4), "\n")
```

```
## Critical Value (tc): 1.6479
```

```
# 5. 結論判斷
if (t_stat_b > t_critical_b) {
  cat("Conclusion: Reject H0. The marginal effect is significantly greater than $13,000.\n")
} else {
  cat("Conclusion: Do not reject H0. No evidence that marginal effect exceeds $13,000.\n")
}
```

```
## Conclusion: Reject H0. The marginal effect is significantly greater than $13,000.
```

Part (c): Expected Price and Interval Estimate (2,000 sqft)

Answer:

- **Point Estimate:** The expected price for a 2,000 sqft house is calculated by substituting $SQFT = 20$ into the estimated quadratic equation: $\hat{PRICE} = \hat{\alpha}_1 + \hat{\alpha}_2(20^2)$.
- **95% Interval Estimate Interpretation:** According to the frequentist definition of confidence intervals, in repeated sampling, approximately 95% of the intervals constructed using this specific methodology will contain the true expected price for a house of this size.
- **Precision:** This interval incorporates the variances of both the intercept and the slope, as well as their covariance, to ensure a precise estimate of the population mean price for 2,000 sqft houses.

```
## c
# Expected Price and 95% Interval for 2000 sqft (SQFT = 20)

# 1. 提取係數與共變異數矩陣
a1_hat <- coef(mod_quadratic)["(Intercept)"]
a2_hat <- coef(mod_quadratic)["I(sqft^2)"]
v_cov_matrix <- vcov(mod_quadratic)

# 2. 點估計值 (Point Estimate)
# 當 SQFT = 20, SQFT^2 = 400
sqft_value <- 20
sqft_sq <- sqft_value^2 # 400
price_expected <- a1_hat + a2_hat * sqft_sq

# 3. 計算期望值的標準誤
#  $Var(a1 + 400*a2) = Var(a1) + 2 * 400 * Cov(a1, a2) + 400^2 * Var(a2)$ 
var_price <- v_cov_matrix["(Intercept)", "(Intercept)"] +
  2 * sqft_sq * v_cov_matrix["(Intercept)", "I(sqft^2)"] +
  (sqft_sq^2) * v_cov_matrix["I(sqft^2)", "I(sqft^2)"]

se_price <- sqrt(var_price)

# 4. 找出 95% t 臨界值 (雙尾)
df_quadratic <- df.residual(mod_quadratic)
t_critical_c <- qt(0.975, df_quadratic)

# 5. 計算信賴區間
lower_c <- price_expected - t_critical_c * se_price
upper_c <- price_expected + t_critical_c * se_price

# 6. 輸出結果
cat("Expected Price for 2000 sqft: $", round(price_expected, 4), " (in $1000)\n", sep="")
```

```
## Expected Price for 2000 sqft: $167.3735 (in $1000)
```

```
cat("95% Confidence Interval: [", round(lower_c, 4), ", ", round(upper_c, 4), "]\n", sep="")
```

```
## 95% Confidence Interval: [158.0481, 176.6988]
```

Part (d): Sample Mean Comparison and Compatibility

Answer:

- **Empirical Observation:** To check the model's local accuracy, we locate houses in the actual dataset that have exactly 2,000 sqft and calculate their average selling price.
- **Compatibility Analysis:** The sample average is considered compatible with the model's prediction if it falls within the 95% confidence interval calculated in Part (c).

- **Conclusion:** While the model represents a global fit across all 500 observations, the sample mean is a direct local average. If the local average is contained within the interval, it suggests the quadratic functional form successfully captures the price behavior at this specific point in the market.

```
## d
```

```
# 1. 篩選所有 SQFT = 20 的房屋資料
```

```
houses_20 <- subset(collegetown, sqft == 20)
```

```
n_houses_20 <- nrow(houses_20) # 列出房屋數量
```

```
# 2. 計算樣本平均售價
```

```
sample_mean_20 <- mean(houses_20$price)
```

```
# 3. 輸出結果
```

```
cat("Number of houses with SQFT = 20: ", n_houses_20, "\n")
```

```
## Number of houses with SQFT = 20: 3
```

```
cat("Sample average price for SQFT = 20: $", round(sample_mean_20, 4), " (in $1000)\n", sep="")
```

```
## Sample average price for SQFT = 20: $163.3333 (in $1000)
```

```
cat("Predicted price from part (c): $", round(price_expected, 4), " (in $1000)\n", sep="")
```

```
## Predicted price from part (c): $167.3735 (in $1000)
```

```
# 4. 檢查相容性 (是否落在區間內)
```

```
is_compatible <- sample_mean_20 >= lower_c && sample_mean_20 <= upper_c
```

```
if (is_compatible) {
```

```
  cat("Conclusion: The sample average is COMPATIBLE with the result in part (c).\n")
```

```
} else {
```

```
  cat("Conclusion: The sample average is NOT compatible with the result in part (c).\n")
```

```
}
```

```
## Conclusion: The sample average is COMPATIBLE with the result in part (c).
```